FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2021

PAPER –1

TEST DATE: 04-07-2021

ANSWERS, HINTS & SOLUTIONS

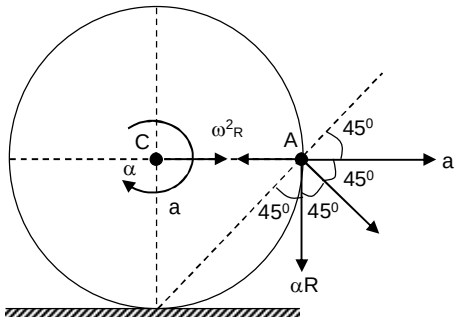
Physics

PART – I

SECTION – A

1. B

Sol. Velocity of point 'A' $v_A = \sqrt{v^2 + \omega^2 R^2} = v\sqrt{2}$



Normal acceleration of point A,

$a_A(n) = \omega^2 R\cos 45^0 + \alpha R\cos 45^0 - a\cos 45^0$

$a_{A(n)} = \dfrac{\omega^2 R}{\sqrt{2}} = \dfrac{v^2}{\sqrt{2}R}$

$\therefore$ radius of curvature of trajectory of point 'A' relative to the ground is

$r = \dfrac{(v_A)^2}{a_{A(n)}} = \dfrac{\left(v\sqrt{2}\right)^2}{\frac{v^2}{\sqrt{2R}}} = 2\sqrt{2}R$

2. A

Sol. $mg\left(r - R\sin\theta\right) = MgR\sin\theta$

$mr = \left(M + m\right)R\sin\theta$

$\sin\theta = \dfrac{mr}{\left(M+m\right)R} = \dfrac{1}{2}$

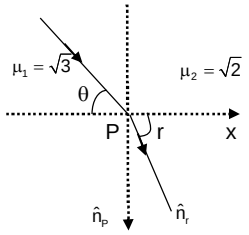
$\therefore \theta = 30^0$

3. B

Sol. At the resonance source voltage and source current in same phase Hence "B"

4. C

Sol. Angle between normal x-axis and incident ray is



$\left(5\sqrt{2}\hat{i} - 5\hat{j} - 5\hat{k}\right).\left(\hat{i}\right) = 10.1.\cos\theta$

$\cos\theta = \dfrac{1}{\sqrt{2}} \Rightarrow \theta = 45^0$

Applying snell's Law at P

$\mu_1 \sin\theta = \mu_2 \sin r$

$\sqrt{3}.\dfrac{1}{\sqrt{2}} = \sqrt{2}\sin r$

$r = 60^0$

The unit vector along the refracted ray

$\hat{n}_r = \cos r\hat{i} + \sin r\hat{n}_P$

$\hat{n}_r = \cos 60^0\hat{i} + \sin 60^0\left(\dfrac{-\hat{j}-\hat{k}}{\sqrt{2}}\right)$

$\hat{n}_r = \dfrac{1}{2}\hat{i} + \dfrac{\sqrt{3}}{2\sqrt{2}}\left(-\hat{j}-\hat{k}\right)$

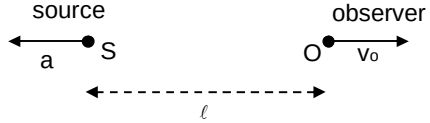
$\hat{n}_r = \dfrac{\sqrt{2}\hat{i} - \sqrt{3}\hat{j} - \sqrt{3}\hat{k}}{\sqrt{8}}$

5. B

Sol. Apparent wavelength

$\lambda' vT + \dfrac{1}{2}aT^2$

Apparent frequency received by the observer

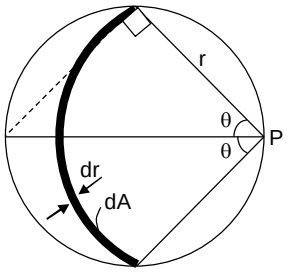


$$f' = \frac{v - v_0}{\lambda'} = \frac{(v - v_0)}{vT + \frac{1}{2}aT^2}$$

$$= \frac{(v - v_0)}{\frac{v}{f} + \frac{a}{2f^2}} = \frac{2(v - v_0)f^2}{(2vf + a)}$$

6. D

Sol. Work done is moving a unit positive charge from infinity to a point on the circumference will be numerically equal to the potential at that point.



So we need to calculate potential at any point on the circumference of this charged disc.

Let choose an element of area dA in the form of a ring of radius r and width dr, where $dA = 2\theta r\, dr$

$r = 2R\cos\theta \qquad \therefore\ dr = -2R\sin\theta d\theta$

Potential at point P due to the charge on this element is

$$dV = -\frac{\sigma R}{\pi\varepsilon_0}\theta\sin\theta d\theta$$

$$\therefore V = \int dV = -\frac{\sigma R}{\pi\varepsilon_0}\int_{\pi/2}^{0}\theta\sin\theta d\theta = -\frac{\sigma R}{\pi\varepsilon_0}\left[-\theta\cos\theta + \sin\theta\right]_{\pi/2}^{0} = \frac{\sigma R}{\pi\varepsilon_0}$$

7. AC

Sol. To find F draw the F.B.D of the mass. then apply work energy theorem to get maximum speed.

8. ABCD

Sol. (A) when S_1, S_2 both are open capacity of the system is C_1, Then C_1.

$$C_1 = \frac{\frac{4\pi\varepsilon_0(4R)(2R)}{(4R-2R)}(4\pi\varepsilon_0 6R)}{\frac{4\pi\varepsilon_0(4R)(2R)}{(4R-2R)} + 4\pi\varepsilon_0 6R}$$

$$C_1 = \frac{48\pi\varepsilon_0 R}{5}$$

(B) When S_1 is closed but S_2 is open capacity is C_2. Find C_2.

$$C_2 = \frac{(48\pi\varepsilon_0 R)(24\pi\varepsilon_0 R)}{(24\pi\varepsilon_0 R + 24\pi\varepsilon_0 R)} = 16\pi\varepsilon_0 R$$

(C) When S_2 is closed but S_2 is open capacity is C_3. Find C_3.

$$C_3 = \frac{(24\pi\varepsilon_0 R)(24\pi\varepsilon_0 R)}{(24\pi\varepsilon_0 R + 24\pi\varepsilon_0 R)} = 12\pi\varepsilon_0 R$$

(D) when both S_1 and S_2 are closed. Capacity is C_4 is. Then C_4 is equal to

$C_4 = 24\pi\varepsilon_0 R$

9. ABCD

Sol. When only S_2 is open

$$\frac{2}{3}=\left(\frac{4}{120}\right)\ell \Rightarrow \ell = 20\text{cm}$$

When only S_1 is open

$$2=\left(\frac{4}{120}\right)\ell \Rightarrow \ell = 60\text{ cm}$$

When all three are closed

$$2=\frac{4}{120}\ell \Rightarrow \ell = 60\text{ cm}$$

When only S_1 is closed

$$2=\frac{4}{120}\times \ell \Rightarrow \ell = 60\text{ cm}$$

10. ABCD

Sol. $\Delta X_{max} = 2.5\,\text{mm}$

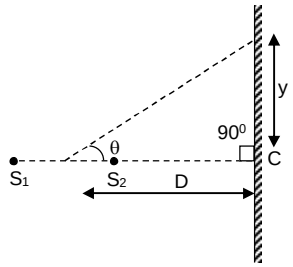
$$n_{max}=\frac{2.5\times 10^{-3}}{5000\times 10^{-10}}=5000$$

For 3000th order bright fringe, n = 3000

$$d\cos\theta = (3000\lambda)$$

$$\cos\theta=\frac{3000\lambda}{5000\lambda}=\frac{3}{5}$$

$$\frac{D}{\sqrt{y^2+D^2}}=\frac{3}{5}$$



Solving $y=\left(\frac{8}{3}\right)\text{m}$

For 5th bright fringe from 'C' n = 4996

$$5000\lambda\cos\theta = 4996\lambda$$

$$\sin\theta = 0.04$$

$$\tan\theta = y/D = 0.04$$

$$y = 8\times 10^{-2}\text{m}$$

For 2500th order bright fringe, n = 2500

$$\cos\theta = 1/2$$

$$\frac{D}{\sqrt{y^2+D^2}}=\frac{1}{2}$$

$$4D^2 = y^2 + D^2$$

$$y=\sqrt{3}D$$

$$y = 2\sqrt{3}\text{ m}$$

11. ABCD

Sol. $C = C_Y + \frac{P}{n}\left(\frac{dV}{dT}\right)$ $\qquad T = \left(\frac{T_0}{V_0}\right)V^2$

$= \frac{3}{2}R + \left(\frac{nRT}{V.n}\right)\left(\frac{V_0^2}{2VT_0}\right)$ $\qquad \frac{dT}{dV} = \frac{T_0}{V_0^2}2V$

$= \frac{3}{2}R + \frac{RTV_0^2}{2T_0V^2}$

$= \frac{3}{2}R + \frac{R}{2} = \frac{4}{2}R = 2R$

$T_f = \left(\frac{T_0}{V_0^2}\right)(2V_0)^2 = 4T_0$

$\Delta Q = nC\Delta T = 1(2R)(4T_0 - T_0) = 6RT_0$

12. ACD

Sol. $m_A = m = 0.3kg$

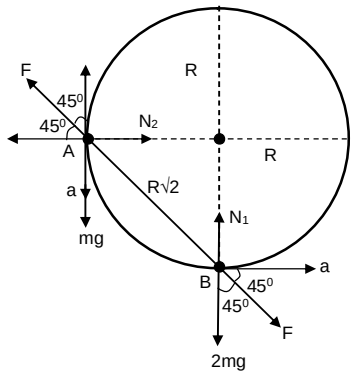
$m_B = 2m = 0.6kg$

$N_1 = 2mg + \frac{F}{\sqrt{2}}$ ...(i)

$\frac{F}{\sqrt{2}} = 2ma$ ...(ii)

$N_2 = \frac{F}{\sqrt{2}}$ ...(iii)

$mg - \frac{F}{\sqrt{2}} = ma$ ...(iv)



From (ii) and (iv)

$mg = 3ma \Rightarrow a = g/3$

from (ii), $\frac{F}{\sqrt{2}} = 2ma = \frac{2mg}{3} = 2$

$\therefore F = 2\sqrt{2}N$

$\therefore N_2 = \frac{F}{\sqrt{2}} = 2N \Rightarrow N_2 = 2N$

$From (i) N_1 = 2mg + \frac{F}{\sqrt{2}} = 2mg + \frac{2mg}{3} = \frac{8mg}{3} = 8N$

$\Rightarrow N_1 = 8N$

SECTION – C

13. 00012.50

Sol. Impulse, $\int F dt = m\sqrt{2gh}$

$\int (i\ell B)dt = m\sqrt{2gh}$ ...(i)

$\int i dt = \Delta q$ ...(ii)

$\Delta q = \dfrac{m\sqrt{2hg}}{B\ell} = \dfrac{20\sqrt{2\times 10\times 5}}{2\times 8} = 12.50$ Coulomb

14. 00014.80

Sol. $\dfrac{dN}{dt} = r - \lambda N$

$\int\limits_{N_0}^{N} \dfrac{dN}{r-\lambda N} = \int\limits_{0}^{t} dt$

$\dfrac{dN}{dt} = r - \lambda N = (r - \lambda N_0)e^{-\lambda t}$

$= \left(10^4 - \dfrac{10^5}{20}\right)e^{-t/20} = \dfrac{5000}{7.4} = \left(\dfrac{10^4}{14.8}\right)$

15. 00002.20

Sol. $E = \sigma T^4$

$\Rightarrow \dfrac{E_2}{E_1} = \dfrac{T_2^4}{T_1^2} = 256 = 4^4$

$\Rightarrow T_2 = 4T_1$

Also, $\lambda_1 T_1 = \lambda_2 T_2$ gives $\lambda_2 = \dfrac{\lambda_1}{4} = 1000\,\overset{0}{A}$

$KE_{max} = \dfrac{hC}{\lambda_2}\phi \Rightarrow \dfrac{12400}{1000} - \phi = 12.4 - \phi$ ...(i)

$\Delta E_{2\to 4} = 13.6\times 2^2 \times \left[\dfrac{1}{2^2} - \dfrac{1}{4^2}\right] = 13.6\times 4\times \dfrac{3}{16} = 10.2\text{eV}$ ...(ii)

From (i) and (ii) $12.4 - \phi = 10.2$

$\Rightarrow \phi = 2.2\,\text{eV}$

16. 00003.25

Sol. $r_{AB} = \int\limits_0^1 \lambda dx = \int\limits_0^1 (1+2x)dx = 2\Omega$

$r_{AC} = \int\limits_0^{0.5} (1+2x)dx = \dfrac{3}{4}\Omega$

Potential difference between X and Y = 4V = potential difference between A and C when $i_G = 0$

$\therefore 4 = \dfrac{28}{(r+2)}\times\dfrac{3}{4} \Rightarrow r = 3.25\Omega$

17. 00012.00

Sol. $v_0 \cos 30^0 = v \cos 45^0$

$\Rightarrow v = v_0\sqrt{\frac{3}{2}}$

Now using conservation of energy

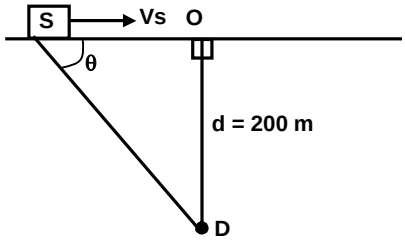
$\frac{1}{2}m\left(v^2 - v_0^2\right) = mgh$

$\frac{3v_0^2}{2} - v_0^2 = 2gh \Rightarrow \frac{v_0^2}{2} = 2gh$

$\Rightarrow v_0 = \sqrt{4gh} = \sqrt{4\times 10\times 3.6} = 12 \text{ m/s}$

18. 00625.00

Sol. $\cos\theta = \frac{v_s t}{vt} = \frac{v_s}{v} = \frac{66}{330} = \frac{1}{5} = 0.2$



$f = \left(\frac{v}{v - v_s\cos\theta}\right)f_0 = \frac{f_0}{\left(1-\frac{1}{25}\right)} = \frac{25}{24}\times 600 = 625 \text{ Hz}$

Chemistry

PART – II

SECTION – A

19. D

Sol. Fact, on heating with concentrated NaOH solution in an inert atmosphere of CO_2, white phosphorus gives PH_3 gas.

20. B

Sol. $\frac{K_t}{K_0} = (\text{Temperature coefficient})^{\Delta t/10}$

$\frac{K_t}{K_0} = (T_C)^{\left(\frac{t-0}{10}\right)}$

$\ln K_t - \ln K_0 = \frac{t}{10}(\ln T_C)$

On comparing

Now, $\frac{t}{10}(\ln T_C) = \frac{t}{10}(\ln 2)$

$T_C = 2$

21. B

Sol. (A) $Be(OH)_2 + 2\overline{O}H \rightarrow \left[Be(OH)_4\right]^{-2}$

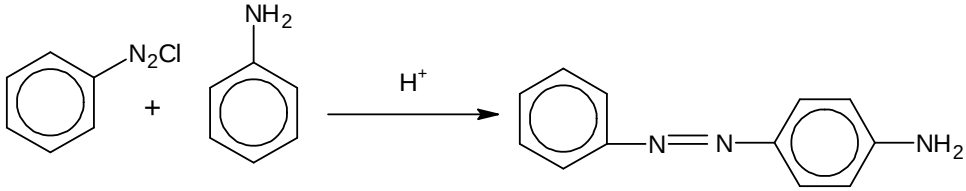
(B) Isoelectric point can be calculated by using relation $PI = \frac{pK_{a_1} + pK_{a_2}}{2}$ and $PI = \frac{pK_{a_2} + pK_{a_3}}{2}$

(C) Fact

(D) $BeF_2 + 2NaF \rightarrow Na_2\left[BeF_4\right]$

22. A

Sol. To stop the unwanted reaction



This reaction is suppressed by adding excess of HCl which will convert free aniline to protonated aniline.

23. A

Sol. $\frac{r_x}{r_y} = \sqrt{\frac{M_y}{M_x}} \Rightarrow \frac{M_y}{M_x} = \frac{2^2}{4^2} = \frac{1}{4}$ .....(i)

$\Rightarrow$ Given that $\frac{W_y}{W_x} = \frac{3}{2}$ .....(ii)

Dividing equation (ii) by (i)

$\frac{W_y}{M_y} \times \frac{M_x}{W_x} = \frac{3}{2} \times \frac{4}{1}$

$\frac{n_y}{n_x} = \frac{6}{1}$

Now $\frac{n_x}{n_y}=\frac{1}{6}\Rightarrow\frac{x_x}{x_y}=\frac{1}{6}$

24. C

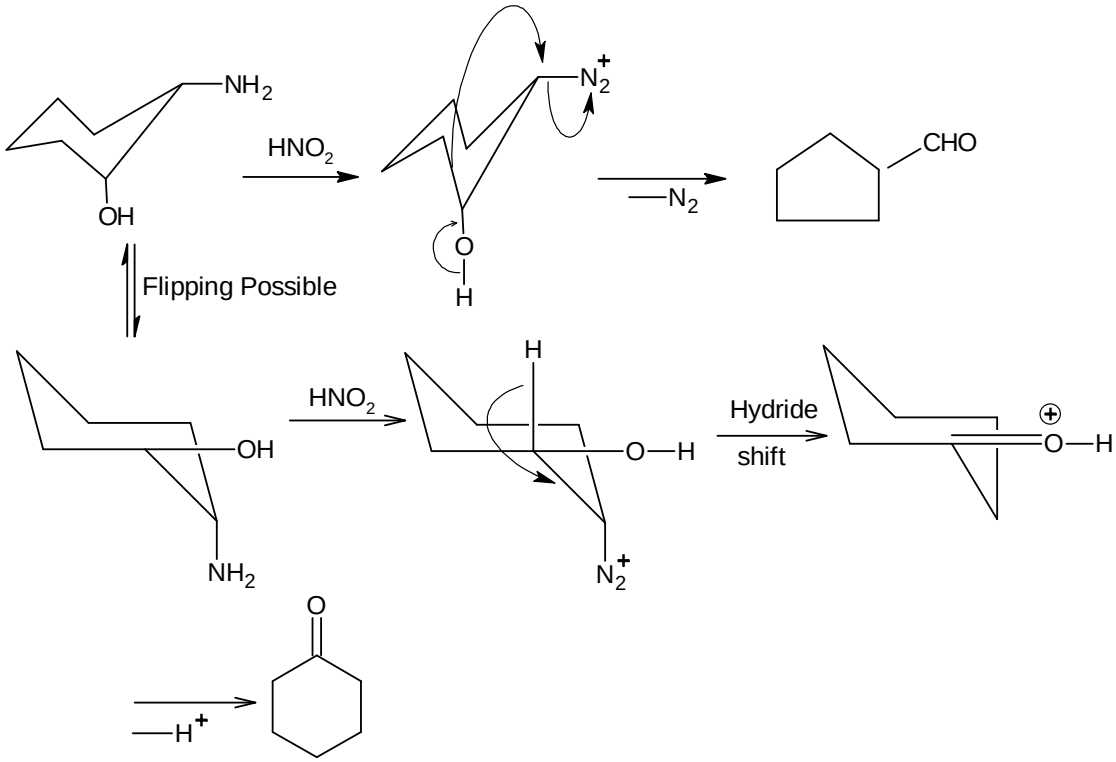
Sol. Room energy $=\frac{3}{2}RT$

$=\frac{3}{2}\times 8.314\times 265$ J/mol

=3.3 kJ mol^{-1}

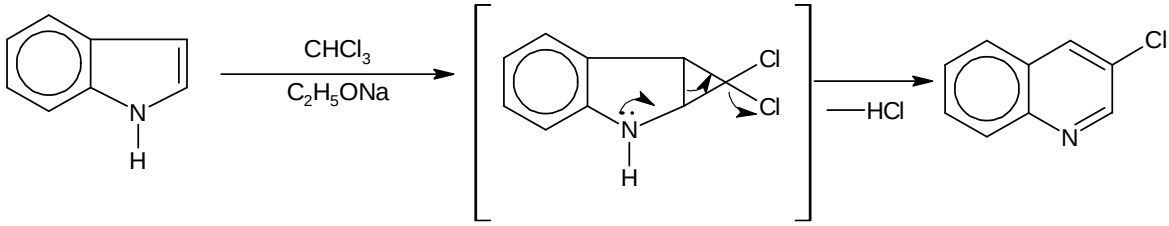
25. ABC

Sol.



26. ABC

Sol.



[$CHCl_3$ + RONa $\longrightarrow$: CCl_2 + R—OH + NaCl]

dichloro methylene

27. BC

Sol. $\underset{x}{SO_2Cl_2}+2H_2O\longrightarrow 2HCl+H_2SO_4$

$$\underset{x}{SO_2Cl_2} + KOH \longrightarrow \underset{y}{2KCl} + \underset{z}{K_2SO_4}$$

$$\underset{y}{KCl} + AgNO_3 \longrightarrow \underset{\text{white ppt.}}{AgCl} + KNO_3$$

$$\underset{z}{K_2SO_4} + BaCl_2 \longrightarrow \underset{\text{white ppt}}{BaSO_4} + 2KCl$$

28. AC

Sol. (i) To show enantiomers molecule must be chiral with or without chiral centers.
(ii) Selective reduction of double bounds in compounds containing other reducible groups except $(\equiv)$, Aromatic nitro groups and acyl halides, can be achieved by Pd.

29. ABCD

Sol. (a) $dE = nC_V dT$, (b) $dH = nC_P dT$

(c) $G = H - TS$, $H = E + PV$

Differentiating we get

$dG = d(E + PV) - d(TS)$

$= dE + PdV + VdP - TdS - SdT.....(i)$

From 1st law dq = dE + PdV $\rightarrow$ work of expansion

and $dS = \frac{dq}{T} \rightarrow$ entropy change

$dq = TdS$

Form eq (i) $dG = dq + VdP - dq - SdT$

$dG = VdP - SdT$ .....(ii)

Equation (ii) at const. pressure

$\left(\frac{dG}{dT}\right)_p = -S$

And at const. Temperature

$\left(\frac{dG}{dP}\right)_T = V$

30. ABCD

Sol. $H_2A + NaOH \longrightarrow \underset{\text{Amphiprotic anionic salt}}{NaHA} + H_2O$

$NaHA + NaOH \longrightarrow Na_2A + H_2O$

As from titration curve $\rightarrow V_2 =$ volume of strong base required at 1st equivalence point i.e point B.

And V_4 = volume of strong base required at 2nd equivalence point i.e. point D. Hence at $V_1 = \frac{V_2}{2}$ and $V_3 = \frac{3V_2}{2}$ indicating 1st half and 2nd half equivalence point.

$\Rightarrow$ Therefore at 1st half equivalence point $\frac{[NaHA]}{[H_2A]} = 1,\ pH = pK_{a_1}$

pH = 4

and similarly, at 2nd half equivalence point at C, $\frac{[Na_2A]}{[NaHA]} = 1,\ pH = pK_{a_2}$

At point D, only salt Na_2A will exist hence pH can be calculated by using salt hydrolysis phenomena.

SECTION – C

31. 00010.50

Sol. $3B_2H_6 + 6NH_3 \rightarrow 3\left[BH_2(NH_3)_2\right]^+\left[BH_4\right]^- = x$

$\downarrow$ heat

$2B_3N_3H_6 + 12H_2$

'y'

Steric factor (f) of $\left[BH_4\right]^- = 4$

$a = \dfrac{2+4+15}{2}$

$= 10.5$

32. 00001.26

Sol. For multielectron species

$I.E = I.E_H \times \dfrac{Z_{eff}^2}{n^2}$, Li = $1s^2 2s^1$

$n = 2$ (for outer electron)

$5.40 = 13.6 \times \dfrac{Z_{eff}^2}{2^2}$

$Z_{eff} = \sqrt{\dfrac{5.40}{13.6}} \times 2$

$Z_{eff} = 1.26$

33. 00272.64

Sol. $\dfrac{\Delta T_b}{\Delta T_f} = \dfrac{K_b}{K_f}$

$= \dfrac{RT_b^2}{1000 \times L_{vap}} \times \dfrac{1000 \times L_{fus}}{RT_f^2}$

$= \dfrac{T_b^2}{T_f^2} \times \dfrac{L_{fus}}{L_{vap}}$

$\dfrac{\Delta T_b}{\Delta T_f} = \dfrac{(373)^2 \times 80}{(273)^2 \times 540} \Rightarrow T_b = 373$ K

$\Rightarrow T_f = 273$ K

$\because \Delta T_b = 0.1$

$\Delta T_f = 0.361$

or $273 - T_f = 0.361$

$T_f = 272.639$ K

34. 00003.92

Sol. By Kohlrausch law,

$\lambda^o_{NH_4OH} = \lambda^o_{NH_4^+} + \lambda^o_{\overline{O}H}$

$\wedge^{o}_{NH_4Cl} + \wedge^{o}_{NaOH} - \wedge^{o}_{NaCl}$

$= 129.8 + 217.4 - 108.9$

$= 238.3 \text{ scm}^2 \text{ mol}^{-1}$

$\wedge_C = 9.33 \text{ scm}^2 \text{ mol}^{-1} (\text{Given})$

$\therefore$ % Degree of dissociation = $\frac{\wedge_C}{\wedge_O} \times 100$

$= \frac{9.33}{238.3} \times 100$

$= 3.92\%$

35. 00027.43

Sol. $\frac{E_2}{E_1 + E_2} = \frac{K_2[\text{tert} - \text{butyl bromide}][\bar{O}H]}{K_2[\text{tert} - \text{butyl bromide}][\bar{O}H] + K_1[\text{tert} - \text{butyl bromide}]}$

$= \frac{7.1 \times 10^{-5} \times 5}{7.1 \times 10^{-5} \times 5 + 1.5 \times 10^{-5}}$

$= 0.96$

$x = 96\%$

Now $\frac{x}{3.5} \Rightarrow \frac{96}{3.5}$

$= 27.428$

36. 00001.26

Sol. Density of Fcc= $\frac{z_1 \times \text{atomic mass}}{N_A \times V_1}$

Density of BCC= $\frac{z_2 \times \text{atomic mass}}{N_A \times V_2}$

$\therefore \frac{d_{fcc}}{d_{Bcc}} = \frac{z_1}{z_2} \times \frac{V_2}{V_1}$

$= \frac{4}{2} \times \frac{(3)^3}{(3.5)^3}$

$= 1.259 = a$

And number of unpaired $\bar{e}$ in $[NiF_6]^{-2}$ is zero as it is low spin complex.

Hence y = 0

Therefore $n = 1.259 + 0$

=1.259

Mathematics

PART – III

SECTION – A

37. B

Sol. Using LMVT in [1, 2]

$$f'(c)=\frac{f(2)-f(1)}{2-1}\le 2\Rightarrow f(2)\le 4$$

Again using in $[2,\ 4]$

$$f'(\alpha)=\frac{f(4)-f(2)}{4-2}\le 2\Rightarrow f(2)\ge 4\ \ \therefore f(2)=4$$

38. A

Sol. $x^2-3x+4>0\Rightarrow x\in R,\ (a-1)x^2-(a+|a-1|+2)x+1\ge 0$

$a-1>0$ and $(a+|a-1|+2)^2-4(a-1)\le 0\Rightarrow a\in\phi$

39. B

Sol. $\because GM\ge H.M\Rightarrow \sqrt{ac}\ge b$

And $\frac{a^{101}+c^{101}}{2}\ge 2b^{101}\Rightarrow$ product of roots is $-$ve.

40. C

Sol. $|a|+|b|=|a+b|\Rightarrow a.b\ge 0$

$(\sin^{-1}x-x).(x^2+5x+6+\cos^{-1}x)\ge 0\Rightarrow \sin^{-1}x\ge x$

$\therefore x\in[0,1]$

41. D

Sol. An odd function

$\therefore$ point $(3,-2)\ \&\ (-5,-4)$ will also lie on the curve for minimum number of roots f(x) has atleast three real roots.

42. C

Sol. $P\left(n,\sqrt{1+n^2}\right)$

$$d_n=\left|\frac{n-\sqrt{n^2+1}}{\sqrt{2}}\right|$$

$$\therefore \lim_{n\to\infty}(n.d_n)=\frac{1}{2\sqrt{2}}$$

43. BCD

Sol. f(x) has minimum and maximum on $[-3,4]$ by IVT.

$f''(x)>0\Rightarrow f(x)$ is concave up.

If f(3) = f(4) then Rolle's theorem $\Rightarrow f'(c)=0$ and $c\in(3,4)$.

44. AC

Sol. Let line $\frac{x-1}{a}=\frac{y-4}{b}=\frac{z-3}{c} \therefore 2a+b+4c=0,\ 3a+2b-2c=0$

$\therefore$ line is $\frac{x-1}{-10}=\frac{y-4}{16}=\frac{z-3}{1}=\lambda \Rightarrow 357=(100+256+1)\lambda^2$

$\therefore \lambda=1 \ \therefore$ points $(-9,20,4)$ or $(11,-12,2)$

Hence $a_1+a_2+a_3=15$ or 1.

45. ABD

Sol. No. of function = 81 No. of onto functions = 30

$\therefore$ No. of functions which are not surjective = 45.

46. CD

Sol. f(x) is discontinuous where $\sin^2 x-\sin x-1=0$

$\therefore \sin x=\frac{1-\sqrt{5}}{2}$

For four points of discontinuity n can be 4 or 5.

47. ABC

Sol. $f(x)=\frac{\sin x}{x}$

48. AD

Sol. $f'(x)=e^x+\frac{1}{x}>0 \Rightarrow f(g(x))=x$

$f'(g(x)).g'(x)=1 \Rightarrow g'(e)=\frac{1}{f'(1)}=\frac{1}{e+1}$

$g''(e)=\frac{-f''(1)}{(f''(1))^3}=\frac{1-e}{(1+e)^3}$

SECTION – C

49. 00005.33

Sol. Let $f'(x)=kx^2(x-2)(1+x^2), k\neq 0 \Rightarrow f'(x)=k(x^5-2x^4+x^3-2x^2)$

$f(x)=\frac{kx^3}{60}(10x^3-24x^2+15x-40) \Rightarrow \frac{f(2)}{f(1)}=\frac{8(-26)}{(-39)}=5.33$

50. 00033.54

Sol. Let $\vec{a}=3\hat{i}+4\hat{j}+10\hat{k},\ \vec{b}=f(x)\hat{i}+g(x)\hat{j}+h(x)\hat{k}$

$\vec{a}.\vec{b}=3f(x)+4g(x)+10\ h(x)=\sqrt{9+16+100}.\sqrt{f^2+g^2+h^2}\ \cos\theta$

$(3f(x)+4g(x)+10\ h(x))_{max}=\sqrt{125}.\sqrt{9}=33.54$

51. 00008.00

Sol. $\because$ injective $\therefore \frac{p}{r}=\frac{p-q}{q-r}$ and $\frac{q}{r}=\frac{r}{p} \therefore$ p,r,q are in G.P

Let $r = p\lambda$, $q = p\lambda^2$ solving $\lambda = -2,1$

$\Rightarrow (p,-2p,4p),(p,p,p)$ Here (p,p,p) is not possible.

$\therefore p - 2p + 4p = 6 \Rightarrow p = 2.$

52. 00000.00

Sol. Let $A = \left[a_{ij}\right]_{3\times3}$, $B = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B^TAB = 0$

$\Rightarrow \left(a_{11}x^2 + a_{22}y^2 + a_{33}z^2\right) + \left(a_{12} + a_{21}\right)xy + \left(a_{13} + a_{31}\right)xz + \left(a_{23} + a_{32}\right)yz = 0$

$\because$ if hold for every column rector B

$\therefore a_{ii} = 0$ and $a_{ij} = -a_{ji} \forall i \neq j$

$\therefore$ A must be skew symmetric

And C is also skew symmetric

$\therefore \det\left(\text{Adj } A\right) + \det\left(\text{Adj } C\right) = 0$

53. 00000.25

Sol. $f'(x) - f(x) = \dfrac{1}{x} - 1 - \ln(x/e) \Rightarrow e^{-x}f'(x) - e^{-x}f(x) = e^{-x}\left(1/x - 1 - \ln(x/e)\right)$

$\therefore f(x) = 1 + \ln(x/e) + \lambda e^x \qquad \because f(1) = e \therefore \lambda = 1$

$\therefore \text{Area} = \int_0^1 x \ln x \, dx = 1/4$

54. 00032.00

Sol. $\left|\int_2^{10} g(x)dx - 8g(\alpha)\right| = \left|\int_2^{10}\left(g(x) - g(\alpha)dx\right)\right| \le \int_2^{10} |\, g(x) - g(\alpha)\,| \, dx \le \int_2^{10} |\, x - \alpha\,| \, dx$

$\therefore \int_2^{10} |\, x - \alpha\,| \, dx = \int_2^{\alpha}(\alpha - n)dn + \int_{\alpha}^{10}(x - \alpha)dn = (\alpha - 6)^2 + 16$

$\therefore$ maximum value is 32.